

OPMIG-08: Security, Masking & Compliance

Series: OPMIG — OpenPipeline Migration | **Notebook:** 8 of 10 | **Created:** December 2025 | **Last Updated:** 07/20/2026

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Learning Objectives

By completing this notebook, you will:

1. Configure masking processors for PII protection
2. Use built-in and custom masking patterns
3. 🌟 **NEW:** Implement GDPR compliance checklist
4. 🌟 **NEW:** Implement HIPAA compliance checklist
5. 🌟 **NEW:** Implement PCI-DSS compliance checklist
6. 🌟 **NEW:** Implement SOC 2 compliance checklist

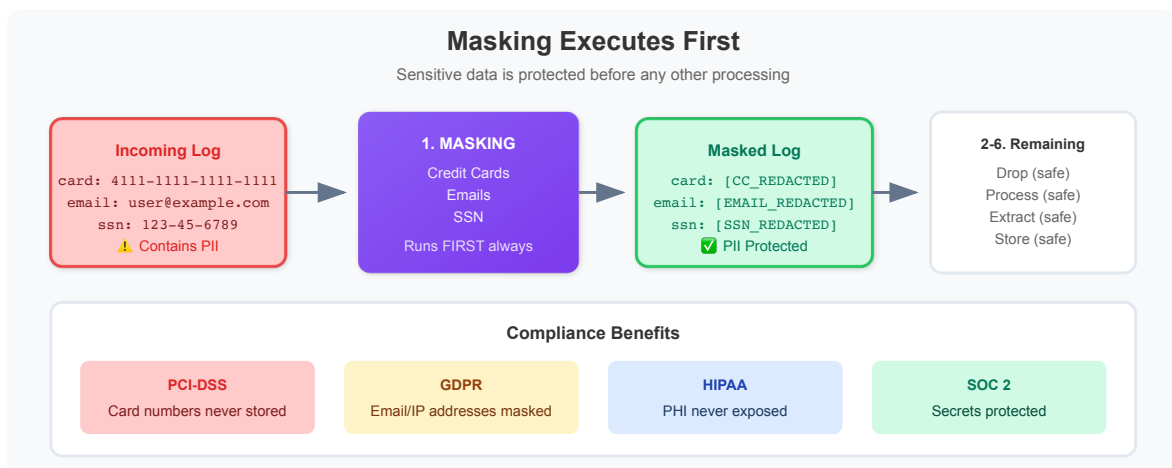
7. Validate masking effectiveness
8. Design complete security pipelines

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS or Managed with Grail and OpenPipeline access
Permissions	<code>openpipeline.configurations.read</code> and <code>openpipeline.configurations.write</code>
API Access	<code>logs.read</code> token scope
DPL Architect	Access to <code>https://{env}.apps.dynatrace.com/ui/apps/dynatrace.dpl.architect</code>
Knowledge	OPMIG-01 through OPMIG-07; understanding of compliance requirements (PCI, HIPAA, GDPR)

Security Stage Overview

Masking executes **first** in the Processing stage, ensuring sensitive data is redacted before any other processing occurs.



Why Masking First?

Order	Reason
Before parsing	Sensitive data never extracted to fields
Before routing	Can't route based on unmasked PII
Before storage	Compliance-safe from the start
Before extraction	Metrics/events don't contain PII

Understanding Masking

What Masking Does

Masking replaces sensitive patterns with redacted values:

Before:

```
User email: john.doe@example.com, card: 4111-1111-1111-1111
```

After:

```
User email: [EMAIL_REDACTED], card: [CC_REDACTED]
```

Masking Processor Configuration

Field	Description
Name	Descriptive name for the mask
Matching Condition	When to apply masking
Fields	Which fields to mask (or all)
Pattern	DPL pattern to match sensitive data
Replacement	Text to replace matches with

Masking vs. Dropping

Action	Use When
Mask	Need the record, just hide sensitive parts
Drop	Entire record is not needed

Built-in Masking Patterns

OpenPipeline includes pre-built matchers for common sensitive data.

Credit Card Numbers

DPL Pattern:

```
CREDITCARD
```

Matches:

- 4111111111111111
- 4111-1111-1111-1111
- 4111 1111 1111 1111

Masking Processor:

```
Name: Mask Credit Cards
Pattern: CREDITCARD:cc
Replacement: [CC_REDACTED]
Fields: content, message
```

Email Addresses

There is no built-in EMAIL matcher in DPL. A pattern of EMAIL is rejected with Named pattern element 'EMAIL' is not valid , so a masking processor built on it never redacts anything. Use the hand-rolled character-class pattern below.

DPL Pattern:

```
[a-zA-Z0-9._%+-]+ '@' [a-zA-Z0-9.-]+
```

Matches:

- user@example.com
- first.last@company.org
- a-b_c%d+e@sub.domain.co.uk

Masking Processor:

```
Name: Mask Emails
Pattern: [a-zA-Z0-9._%+-]+ '@' [a-zA-Z0-9.-]+
Replacement: [EMAIL_REDACTED]
Fields: content, message
```

Note: the domain class includes `.`, so a sentence-final period is absorbed into the match (mail x@y.io. → mail [EMAIL_REDACTED]). That over-masks by one character rather than under-masking — the safe direction for PII.

IP Addresses

DPL Pattern:

```
IPADDR
IPV4ADDR
IPV6ADDR
```

Masking Processor:

```
Name: Mask IP Addresses
Pattern: IPADDR:ip
Replacement: [IP_REDACTED]
Fields: content, client_ip, remote_addr
```

Phone Numbers (Custom Pattern)

DPL Pattern:

```
'(' [0-9]{3} ')' SPACE? [0-9]{3} '-' [0-9]{4}
```

INT does not accept a {n} quantifier — it supports only `*` and `+`, and `INT{3}` fails with `INT` only supports `'*'` (nullable) and `'+'` (not nullable) quantifiers .

Use a `[0-9]{n}` character class for fixed-width digit runs.

Matches:

- (555) 123-4567
- (555)123-4567

Custom Masking with replacePattern

The `replacePattern` function enables custom masking using DPL patterns.

Syntax

```
fieldsAdd content = replacePattern(content, "PATTERN", replacement: "REPLACEMENT")
```

Example: Mask SSN

Pattern: SSN format 123-45-6789

```
fieldsAdd content = replacePattern(content, "[0-9]{3} '-' [0-9]{2} '-' [0-9]{4}", replacement: "[SSN_REDACTED]")
```

Example: Mask API Keys

Pattern: Keys starting with `key_`, masked through to the next whitespace

```
fieldsAdd content = replacePattern(content, "'key_' NSPACE", replacement: "[API_KEY_REDACTED]")
```

Why not 'key_' WORD ? WORD matches a greedy run of alphanumerics and underscores, but stops at `-` . Against `key_XY-99` it yields `[API_KEY_REDACTED]-99` , leaving part of the secret in the log. `NSPACE` consumes to the next whitespace and masks the whole token.

Example: Mask Bearer Tokens

Pattern: Bearer tokens in Authorization headers

```
fieldsAdd content = replacePattern(content, "'Bearer ' NSPACE", replacement: "Bearer [TOKEN_REDACTED]")
```

Example: Mask Passwords in URLs

Pattern: Password parameter in query strings

```
fieldsAdd content = replacePattern(content, "'password=' LD:pwd ('&'|EOL)", replacement: "password=[REDACTED]&")
```

Chaining Multiple Masks

```
// Apply multiple masks in sequence
fieldsAdd content = replacePattern(content, "CREDITCARD", replacement: "[CC_REDACTED]")
| fieldsAdd content = replacePattern(content, "[a-zA-Z0-9._%+-]+ '@' [a-zA-Z0-9.-]+", replacement: "[EMAIL_REDACTED]")
| fieldsAdd content = replacePattern(content, "IPADDR", replacement: "[IP_REDACTED]")
```

Compliance Patterns

PCI-DSS Compliance

Requirements:

- Mask all Primary Account Numbers (PANs)
- Mask CVV/CVC codes
- Mask cardholder names when combined with PANs

Pipeline Configuration:

```
// Mask credit card numbers
fieldsAdd content = replacePattern(content, "CREDITCARD", replacement: "[PAN_REDACTED]")
// Mask CVV (3-4 digits after 'cvv=' or 'cvc=')
| fieldsAdd content = replacePattern(content, "('cvv='|'cvc='|'CVV='|'CVC=')[0-9]{3,4}", replacement: "cvv=[REDACTED]")
```

GDPR Compliance

Requirements:

- Mask personal identifiers (emails, phone numbers)
- Mask IP addresses (considered PII in EU)
- Mask names when identifiable

Pipeline Configuration:

```
// Mask emails
fieldsAdd content = replacePattern(content, "[a-zA-Z0-9._%+-]+ '@' [a-zA-Z0-9.-]+", replacement: "[PII_REDACTED]")
// Mask IP addresses
| fieldsAdd content = replacePattern(content, "IPADDR", replacement: "[IP_REDACTED]")
// Mask user IDs
| fieldsAdd content = replacePattern(content, "'userId=' LD:uid", replacement: "userId=[USER_REDACTED]")
```

HIPAA Compliance

Requirements:

- Mask Protected Health Information (PHI)

- Mask patient identifiers
- Mask medical record numbers

Pipeline Configuration:

```
// Mask patient IDs
fieldsAdd content = replacePattern(content, "'patientId=' LD:pid",
replacement: "patientId=[PHI_REDACTED]")
// Mask MRNs
| fieldsAdd content = replacePattern(content, "'mrn=' LD:mrn", replacement:
"mrn=[PHI_REDACTED]")
// Mask SSNs
| fieldsAdd content = replacePattern(content, "[0-9]{3} '-' [0-9]{2} '-' [0-9]
{4}", replacement: "[SSN_REDACTED]")
```

SOC 2 Compliance

Requirements:

- Mask credentials and secrets
- Mask authentication tokens
- Audit access to sensitive data

Pipeline Configuration:

```
// Mask passwords
fieldsAdd content = replacePattern(content, "('password='|'pwd='|'passwd=')
LD:pwd", replacement: "password=[REDACTED]")
// Mask API keys
| fieldsAdd content = replacePattern(content, "
('api_key='|'apiKey='|'API_KEY=') LD:key", replacement: "api_key=[REDACTED]")
// Mask tokens
| fieldsAdd content = replacePattern(content, "'Bearer ' NSPACE", replacement:
"Bearer [REDACTED]")
```

Field Removal for Security

Sometimes it's better to remove entire fields rather than mask values.

Using fieldsRemove

```
// Remove sensitive fields entirely
fieldsRemove password, secret, api_key, token, authorization
```

When to Remove vs. Mask

Scenario	Action
Field always contains sensitive data	Remove
Only some values are sensitive	Mask
Need to know field existed	Mask with placeholder
Compliance requires no trace	Remove

Conditional Field Removal

Apply removal only when field contains sensitive patterns:

```
// Remove field only if it contains a pattern
fieldsAdd authorization = if(contains(authorization, "Bearer"), null, else:
authorization)
```

Validating Masking

After configuring masking, verify it's working correctly.

 **Important:** Test with sample data before deploying to production.

```
// Check for any remaining credit card patterns in stored logs
// If masking works, this should return 0 results
fetch logs, from: now() - 24h
| filter matchesPhrase(content, "4111")
  OR matchesPhrase(content, "5500")
  OR matchesPhrase(content, "3782")
| limit 10
```

```
// Check for any remaining email patterns
// Look for @ symbol with surrounding text
fetch logs, from: now() - 24h
| filter matchesPhrase(content, "@gmail.com")
  OR matchesPhrase(content, "@yahoo.com")
  OR matchesPhrase(content, "@example.com")
| limit 10
```

```
// Verify redaction placeholders are present
// This confirms masking is actively working
fetch logs, from: now() - 24h
| filter contains(content, "[REDACTED]")
  OR contains(content, "[CC_REDACTED]")
  OR contains(content, "[EMAIL_REDACTED]")
  OR contains(content, "[PII_REDACTED]")
| summarize {masked_count = count()}, by: {dt.openpipeline.pipelines}
| sort masked_count desc
```

```
// Sample masked logs to verify format
fetch logs, from: now() - 24h
| filter contains(content, "REDACTED")
| fields timestamp, content
| limit 20
```

```
// Audit: Count masked records by pipeline and type
fetch logs, from: now() - 24h
| fieldsAdd has_cc_mask = contains(content, "[CC_REDACTED]"),
           has_email_mask = contains(content, "[EMAIL_REDACTED]"),
           has_ip_mask = contains(content, "[IP_REDACTED]"),
           has_pii_mask = contains(content, "[PII_REDACTED]")
| summarize {
    total = count(),
    cc_masked = countIf(has_cc_mask),
    email_masked = countIf(has_email_mask),
    ip_masked = countIf(has_ip_mask),
    pii_masked = countIf(has_pii_mask)
}, by: {dt.openpipeline.pipelines}
```

```
// Check for potential SSN patterns that might be missed
// Pattern: ###-##-#### where # is a digit
fetch logs, from: now() - 24h
| filter matchesPhrase(content, "-")
| filter NOT contains(content, "[SSN_REDACTED]")
| fields content
| limit 50
```

Testing Masking in DPL Architect

Before deploying masking rules, test them:

1. Open **DPL Architect** in Dynatrace
2. Paste sample log with sensitive data
3. Test your DPL pattern matches correctly
4. Verify replacement produces expected output

Test Cases to Include

Test Case	Input	Expected Output
Credit card	4111-1111-1111-1111	[CC_REDACTED]
Credit card no dashes	4111111111111111	[CC_REDACTED]
Email	user@example.com	[EMAIL_REDACTED]

Test Case	Input	Expected Output
IPv4	192.168.1.1	[IP_REDACTED]
IPv6	2001:db8::1	[IP_REDACTED]
Mixed content	Multiple patterns	All redacted

Best Practices

Masking Configuration

Practice	Reason
Mask before routing	Routing rules shouldn't see PII
Use descriptive placeholders	Aids troubleshooting
Test patterns thoroughly	Avoid over/under masking
Document masking rules	Compliance audits

Pattern Design

Practice	Reason
Be specific	Avoid false positives
Handle variations	Same data, different formats
Use built-in matchers	Pre-tested, reliable
Chain patterns	Cover all sensitive data

Compliance

Practice	Reason
Document all masking	Audit trail
Regular pattern review	New data sources

Practice	Reason
Test with real samples	Ensure effectiveness
Monitor for leaks	Continuous validation

Performance

Practice	Reason
Apply to specific fields	Faster than all fields
Use matching conditions	Skip unneeded records
Order patterns by frequency	Most common first

Complete Security Pipeline Example

Pipeline: `payment-logs-secure`

Masking Processor 1: Credit Cards

```
Name: Mask Credit Cards
Pattern: CREDITCARD
Replacement: [CC_REDACTED]
Fields: content, card_number
Matching: (all records)
```

Masking Processor 2: CVV Codes

```
fieldsAdd content = replacePattern(content, "('cvv='|'cvc=') [0-9]{3,4}",
replacement: "cvv=[REDACTED]")
```

Masking Processor 3: Emails

```
Name: Mask Emails
Pattern: [a-zA-Z0-9._%+-]+ '@' [a-zA-Z0-9.-]+
Replacement: [EMAIL_REDACTED]
Fields: content, customer_email
Matching: (all records)
```

Masking Processor 4: IP Addresses

```
Name: Mask IPs
Pattern: IPADDR
Replacement: [IP_REDACTED]
Fields: content, client_ip, remote_addr
Matching: (all records)
```

Masking Processor 5: Auth Tokens

```
fieldsAdd content = replacePattern(content, "'Bearer ' NSPACE", replacement:
"Bearer [TOKEN_REDACTED]")
| fieldsRemove authorization, auth_token
```

Pipeline Verification Query

```
// Verify complete masking for payment-logs pipeline
fetch logs, from: now() - 1h
| filter dt.openpipeline.pipelines == "payment-logs-secure"
| summarize {
    total = count(),
    with_cc = countIf(contains(content, "[CC_REDACTED]")),
    with_email = countIf(contains(content, "[EMAIL_REDACTED]")),
    with_ip = countIf(contains(content, "[IP_REDACTED]")),
    with_token = countIf(contains(content, "[TOKEN_REDACTED]"))
}
```

Next Steps

Now that security is configured, complete your migration:

Notebook	Focus Area
OPMIG-09	Troubleshooting & Validation

References

- [OpenPipeline Masking](#)
 - [DPL replacePattern](#)
 - [Data Privacy in Dynatrace](#)
 - [Compliance Best Practices](#)
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Last Updated: May 6, 2026

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